Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/13/2018
PI Name:	Simpson, Richard Ph.D.		
Project Title:	The Impact of Modeled Microgravity and Prior Radiation Exposure on Cytomegalovirus Reactivation and Host Immune Evasion		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	10/21/2015	**End Date:**	09/30/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	6	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 3/4/19) NOTE: End date changed to 10/20/2018 per NSSC information (Ed., 3/5/18) NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Laughlin, Mitzi Ph.D. (University of Houston) Markofski, Melissa Ph.D. (University of Houston) Mehta, Satish Ph.D. (NASA Johnson Space Center) Pierson, Duane Ph.D. (NASA Johnson Space Center) Rezvani, Katayoun M.D., Ph.D. (The University of Texas MD Anderson Cancer Cente)		
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Performance Goal No.:	
Performance Goal Text:	
Task Description:	Cytomegalovirus (CMV), which infects 50-80% of the US population, and CMV DNA is frequently found in the body fluids of astronauts during space travel indicating that the spaceflight environment can initiate active CMV infections. The effects of microgravity on CMV replication and its potential to evade the host immune system are unknown. In this study, we will utilize the rotating wall vessel (RWV) cell culture system to simulate microgravity on Earth allowing us to study the virulent and infectivity properties of CMV and how it interacts with host immune cells in the absence of gravity. The study hypothesizes that modeled microgravity will evoke changes in cellular gene expression that will lead to CMV reactivation and increase virus replication and infectivity. The researchers believe that modeled microgravity will impair the anti-CMV responses of host T-cells and NK-cells that will allow viral persistence and replication to occur. During testing both traditional and cutting-edge laboratory techniques that are typically used to assess virulence and immune cell function in the transplant patient setting allowing determination if CMV infection in the microgravity environment is likely to increase the risk of an adverse health event during exploration class spaceflight missions. Specific Aims: To determine the impact of simulated microgravity on CMV replication and its potential to evade the host immune system a) To determine the effects of modeled microgravity on CMV replication and the expression of genes associated with latency and lytic activity. b) To determine the impact of modeled microgravity on the anti-CMV activity of host T-cells and NK-cells This study will provide preliminary data to determine if CMV virus displays increased infectivity when exposed to a low shear modeled microgravity environment. Data will be evaluated and added to the evidence base capturing the responses to microorganisms grown in rotating wall vessels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our early results indicate that simulated microgravity might actually improve the ability of T-cells to expand when stimulated with cytomegalovirus peptides. This could have important implications for immunotherapy that rely on methods to increase the number of T-cells reacting to specific virus and tumor antigens so that they may be given to immunocompromised patients after stem cell and solid organ transplantation. Preliminary results indicate that simulated microgravity affects both the ability of cells to become infected with cytomegalovirus and for the virus to replicate during prolonged cultures. These findings add knowledge to the biological basis of this virus, which infects the vast majority of the world's population.
	Project Summary The reactivation of latent herpesviruses poses a significant risk to the health and safety of the crew during exploration class spaceflight missions. Of particular concern is cytomegalovirus (CMV), which infects 50-80% of the US population and is known to cause significant morbidity and mortality in immunocompromised patients. CMV DNA is frequently found in the body fluids of astronauts during space travel indicating that the spaceflight environment can initiate active CMV infections. As spaceflight also causes huge alterations in the normal functioning of the immune system, CMV reactivation in a potentially immunosuppressed crewmember could have disastrous consequences during long-duration exploration class missions. The effects of microgravity on CMV replication and its potential to evade the host immune system are unknown. In this study, we will utilize the rotating wall vessel (RWV) cell culture system to simulate microgravity on Earth allowing us to study the virulent and infectivity properties of CMV and how it interacts with host immune cells in the absence of gravity. We hypothesize that modeled microgravity will evoke changes in cellular gene expression that will underpin CMV reactivation and increase virus replication and infectivity. We further postulate that modeled microgravity will impair the anti-CMV responses of host T-cells and NK-cells that will allow viral persistence and replication to occur. We will use both traditional and cutting-edge laboratory techniques that are typically used to assess virulence and immune cell function in the transplant patient setting. This will allow us to determine if CMV infection in the microgravity environment is likely to increase the risk of an adverse health event during exploration class spaceflight missions. Studies like these are required so that effective countermeasures can be developed that would allow CMV-infected crewmembers to embark on such missions. Findings/Progress to Date: Progress since the last reporting date has been slow, largely due to the PI changing institutions and moving his laboratory from the University of Houston to the University of Arizona. A no-cost extension has been requested to accommodate this. We expect to complete the project within the next 12-months. Below is a summary of the progress/findings made to date for each specific aim of the project: Specific Aim 1: Determine the impact of simulated microgravity on CMV replication and its potential to evade the host immune system Our hypothesis was that, compared to traditional cell culture, simulated microgravity (SMG) will increase lytic gene expression in CMV-infected monocytes and amplify viral replication and infectivity upon monocyte differentiation. Before moving our experiments to primary human monocytes, we deemed it important to first of all establish the CMV infection model using a suitable cell line. We have established a positive infection model for Kasumi-3 myeloid progenitor cells using a low fibroblast passaged, BAC-derived TB40/E clinical strain of CMV. The virus is cell associated as confirmed by both real-time PCR and confocal imaging. Cultures are separated into cell pellet and supernatant fractions via high sped centrifugation, 14,000 x g, and the viral copy number is highest in the cell pellet fraction; confocal imaging confirms internalization of the virus. Lytic infection has been confirmed by direct single color antibody staining using mouse anti-human mab810X-AlexaFluor-488(IE-1) and flow cytometry. Additionally, Kasumi-3 cells pre-treated to 12 days of simulated microgravity (SMG) are seemingly resistant to initial infection at the early 12-24 hour time point, with viral copy numbers being consistently lower than non-treated controls. Further, viral copy numbers are lower in models where non-pretreated Kasumi-3 cells are first infected and then placed into SMG than

Task Progress:	those cells that are infected and maintained in 1g flasks. In our current infection model, Kasumi-3 cells are pre-treated to 12 days of SMG before they are infected at an MOI of 1, using high speed centrifugation (14,000 x g), and then they are monitored over 25 days. The cells undergo 13 days of 1 g culture, and then selected cultures are returned to SMG for 12 days. While we have shown that SMG affects lytic CMV replication, optimizing the model to show that the virus has become latent in Kasumi-3 cells is a major objective for this coming year. This is required to show whether or not SMG can reactivate the virus from a state of latency. In our initial experiments, we have found that reintroducing infected kasumi-3 cells to SMG at day 17 post infection does not result in an increase in detectable CMV, indicating, perhaps, that SMG does not trigger CMV reactivation in the absence of another stimulus. We will, therefore, attempt to reactivate CMV from the infected Kasumi-3 cells by stimulating them with PMA-TPA and quantifying viral load in the cell supernatants and cell pellets as part of a time series experiment. When this has been established we will introduce SMG to the TPA-induced CMV reactivation model. Currently, transcriptional analysis of lytic transcript UL123 is underway to evaluate whether the virus becomes latent over time in culture. Additionally, at the end of the traditional 25-day culture period, (12-O-Tetradecanoylphorbol-13-Acetate) TPA was added to each culture to assess whether Kasumi-3 cell differentiation will affect viral load in either SMG or 1g conditions. This analysis is also underway and data for the TPA treated cultures, as well as the other aliquots from days 0 to 25 will be available shortly. Specific Aim 1 Summary: • We have established a positive infection model for Kasumi-3 cells, and have confirmed infection by real –time PCR as well as confocal imaging. • We have shown that the virus is cell associated. The viral load (copy#/uL) is always highest in the cell pellet fraction. The fractions are separated by high-speed centrifugation, 14,000 x g x 4 minutes, and then the supernatant is removed and evaluated separately from the pelleted fraction. In addition, confocal imaging confirms that the virus is internalized into the kasumi-3 cell. • We have seen that pre-treatment of the Kasumi-3 cells in simulated microgravity seemingly protects the cells against initial infection in the early 12-24 culture period. The cells are infected with a BAC-derived TB40/E clinical strain of CMV at an MOI of 1 using high speed centrifugation, 1000 x g x 30 minutes. • We have seen that for infected Kasumi cells in SMG viral expansion (viral load) is less than infected cells cultured in 1g conditions. So, viral expansion/dissemination/infection is impeded in SMG. • We have seen that direct CMV IE-1 antibody staining with mab810-AlexaFluor 488 is positive by flow cytometry for the very early infection time points 12-24 hours. However, this assay has proven difficult to optimize, and does not speak directly to the establishment of latency. Specific Aim 2: Determine the effects of modeled microgravity on CMV replication and the expression of genes associated with latency and lytic activity. All Kasumi-3 cells from the CMV infection experiments (with and without SMG) have been cryopreserved for future DNA extraction and gene expression assays by RT-PcR. We expect that any increase in CMV reactivation due to SMG or otherwise will be preceded by an expression of lytic genes and a downregulation of latent genes. We also expect to see latent gene expression increase following primary infection to indicate that the virus has become latent. Downregulated IE-1 expression should occur concomitantly with this. Specific Aim 3: determine the impact of modeled microgravity on the anti-CMV activity of host T-cells and NK-cells. In the process of understanding the mechanistic underpinnings of suppression in NK cell function seen in our International Space Station (ISS) study, we have established an in vitro model using a rotary cell culture system to simulate microgravity in human peripheral blood mononuclear cells. Using this model, we were able to show that microgravity adversely affects immune cell viability (<90% viable) after a 12-hour exposure to microgravity. We have also replicated the suppression in NK cell function seen in crewmembers. NK cells exposed to microgravity (SMG) killed fewer cancer cells compared to NK cells exposed to both 1g (static) and a rotational control (RC) (for shear stress). This suppression in function was evident despite absence of any phenotypic differences among the three groups (SMG, static, and RC). Microgravity reduced NK cell perforin expression but did not affect their conjugation rate to target cells. Further, we used cytometry-by-time-of-flight to look at various functional and maturation markers, which hinted at reduced degranulation and cytokine production in NK cells exposed to microgravity. We confirmed this using flow cytometry, discovering that microgravity lowered degranulation of cytotoxic granules (reduced CD107a+ expression) and suppressed TNF-a and IFN-gamma expression in NK cells that were exposed to cancer cells. A manuscript containing this work is currently being prepared for publication in a scientific journal. The remainder of this aim will be spent determining the effects of SMG on the anti-CMV activity of both NK-cells and T-cells. The effects of pre-exposure to SMG on gamma d-T-cell expansion (to zoledronic acid and IL-2) and anti-tumor function (against a range of tumor target cells) will be determined. Similarly, we will determine the effects of SMG on the ex vivo expansion and cytotoxicity of the so-called ‘CMV-specific NK-cells,' defined by their NKG2C+/NKG2A- surface phenotype. The effects of SMG on the ability of these cells to form complexes with CMV-infected target cells (using peptide pulsed autologous PHA blasts as a model) will be determined. We have shown that pre-exposure to SMG, somewhat surprisingly, augments the ex vivo expansion of CMV specific T-cells. Upcoming experiments will determine the functional status of these cells and whether or not expanded CMV-specific T-cells, following exposure to SMG, retain their ability to recognize viral antigens and perform MHC-restricted and antigen-specific cytotoxicity.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
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